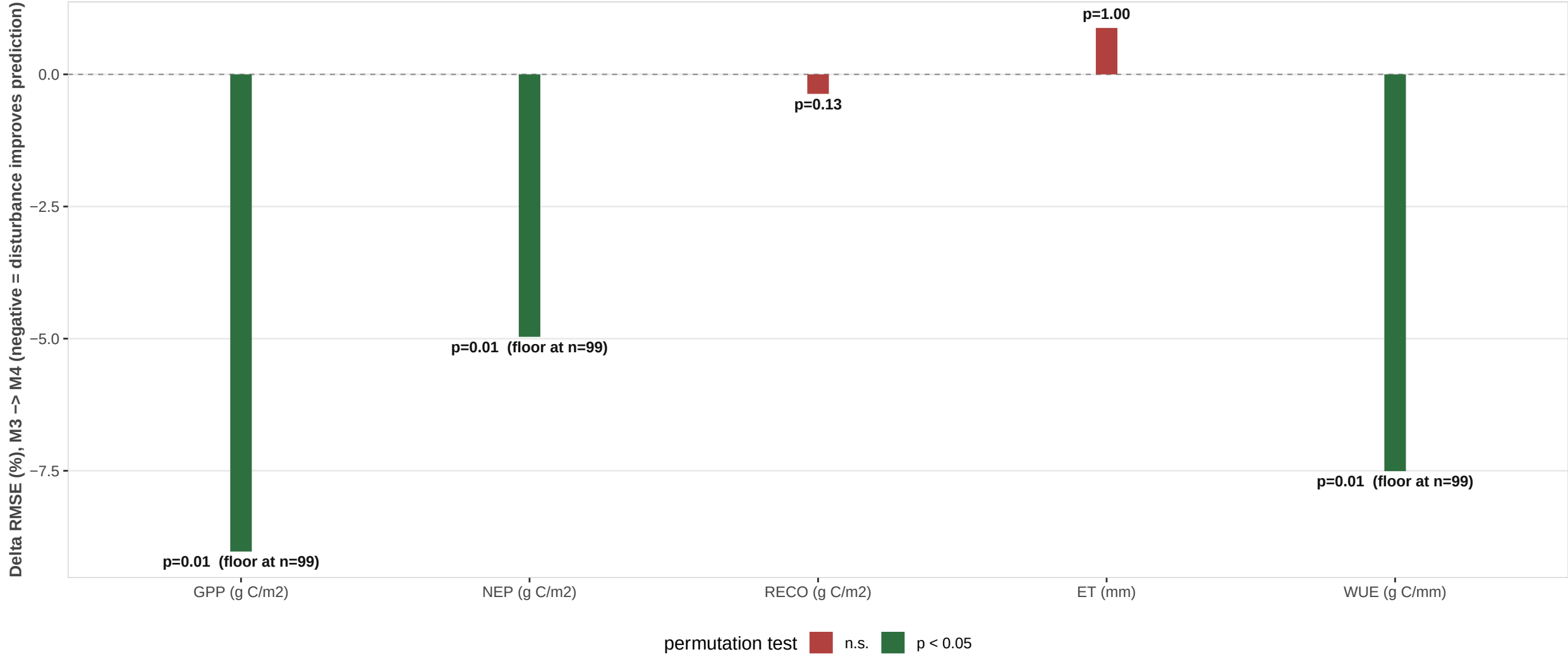
M3 −> M4: does disturbance improve growing−season accumulated flux totals?

Circularity−free response (see daily_reports/2026−09−17−efp−calculation−and−circularity.md) | tc>=30, 93 sites, n=378 site−years
XGBoost (GPPsat's tuned hyperparameters, reused as a first pass), LOSO CV, climate lag1+lag2 + traits + disturbance



p from 99−rep block permutation of the disturbance block. GPP/NEP/WUE match the EFP−based result's significance pattern (GPPsat/NEPmax/WUE real, ETmax null) with smaller magnitudes; RECO joins ET as a null. NEP's M3 R2 is negative (−0.075) − a small residual of two much larger fluxes, hard to predict at all.